

Q.22 Find the value of $\lim_{x \rightarrow 0} \frac{2^x - 3^x}{x}$

Section-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain four laws of thermodynamics.

Q.24 a) Find reverberation time for a hall of dimensions 3000m³ and 150m³ area having average absorption coefficient of 0.2 O.W.U.

b) Explain illumination and efficiency of light.

Q.25 Evaluate $\int x^2 e^x dx$.

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1st Year / Arch. Engg.

Subject : Applied Science & Mathematics

Time : 3 Hrs.

M.M. : 60

Section-A

Note: Multiple Choice questions. All questions are compulsory. (6x1=6)

Q.1 The principle of heat engine is based on :

- (a) Newton's second law
- (b) First law of thermodynamics
- (c) Second law of thermodynamics
- (d) Law of conservation of mass

Q.2 SI unit of radiant flux is:

- (a) Candela
- (b) LUX
- (c) Watt
- (d) Lumen

Q.3 Reverberation time depends on:

- (a) Volume of hall
- (b) Surface area of walls
- (c) Absorption properties of surfaces
- (d) All of the above

Q.4 $\sin (A+B) =$

- (a) $\sin A \sin B + \cos A \cos B$
- (b) $\cos A \cos B - \sin A \sin B$
- (c) $\sin A \cos B + \cos A \cos B$
- (d) $\sin A \cos B - \cos A \cos B$

Q.5 $\frac{d}{dx}(\cos x) =$

- (a) $\sin x$ (b) $-\sin x$
- (c) 1 (d) 0

Q.6 $\int \cos x \, dx =$

- (a) $\sin x$ (b) $-\sin x$
- (c) 0 (d) 1

Section-B

Note: Objective/Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define acoustics.

Q.8 The force of attraction between molecules of different substances is called _____.

Q.9 A solar cell is a device that converts the energy of light directly into _____ by the photovoltaic effect.

Q.10 Evaluate $\sin 105^\circ$.

Q.11 Differentiate $x^2 e^x$ w.r.t. x .

Q.12 Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$

Section-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 State any two methods to control humidity.

Q.14 Give any four methods of water proofing.

Q.15 Differentiate between cohesive and adhesive forces.

Q.16 Define reverberation? State any two methods to control reverberation.

Q.17 Prove that $\frac{\sin A + \sin 3A}{\cos A + \cos 3A} = \tan 2A$

Q.18 Prove that $\tan 9A - \tan 5A - \tan 4A = \tan 9A \tan 5A \tan 4A$.

Q.19 Find the area under the curve $y=x^2$ from $x=0$ to $x=2$.

Q.20 Find derivative of $\frac{\cos x}{x^2}$ w.r.t. x .

Q.21 Find the value of $\int_0^{\frac{\pi}{2}} \cos^5 x \, dx$